

TECHNICAL APPROACH

Technology Stack

Zero-Hardware

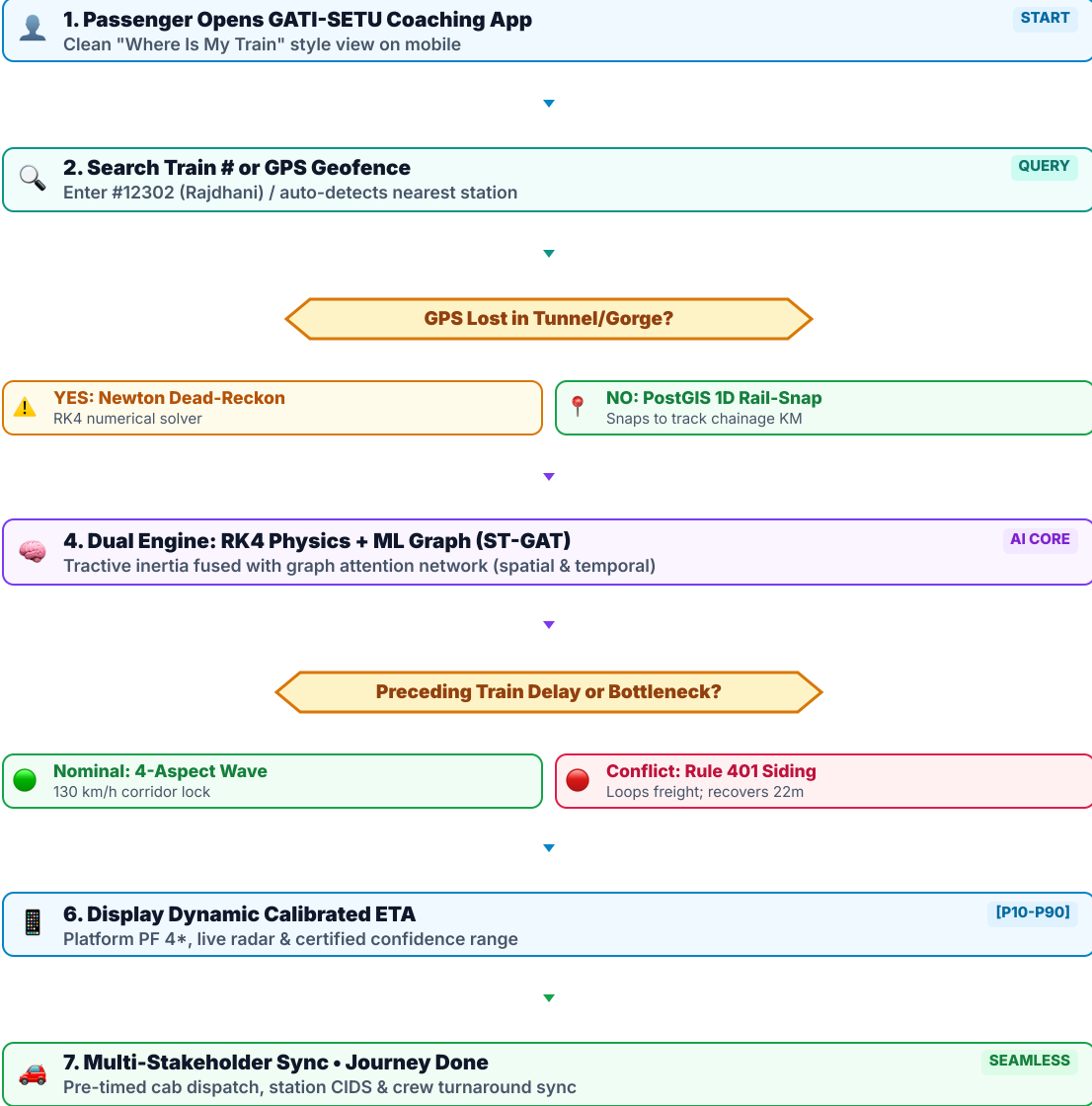
Mobile & Frontend	React Native / React 19, Tailwind CSS v4, Lucide-React, MapLibre GL, Sonner
Backend & Live APIs	Python 3.11, FastAPI (Async), Uvicorn, WebSockets (WSS), Node.js Gateway
Machine Learning (ML) & GNN	PyTorch 2.5, PyTorch Geometric (PyG 2.6) ST-GAT, Scikit-Learn, ONNX Runtime
Statistical Forecasting	Inductive Conformal Prediction [P10–P90], Temporal Quantile Bounds
Physics & Kinematics	Newton-Davis Equation, Runge-Kutta 4th-Order (RK4) ODE Solver
Spatial & Time-Series DB	PostgreSQL 16, PostGIS (1D LRS Chainage KM), TimescaleDB
Streaming & Scalability	Apache Kafka (12,000+ Trains Streamed), Redis 7.2 (Pub/Sub across IR Zones)
Statutory Data Ingestion	BEL RTIS (ISRO NavIC 30s GPS), CRIS COA & FOIS, e-Caution (TSR), Open-Meteo
DevOps & Security	Docker, Kubernetes (K8s), OAuth 2.0 / JWT, AES-256 GCM Encryption

100% ZERO-HARDWARE Taps statutory NavIC & CRIS feeds. Scalable to thousands of coaching trains across all IR zones.



End-to-End User & System Flow

Traveler Journey



Working Prototype & Capabilities

Verified Live



Live Passenger App

Physics Kinematics & 1D Track LRS
Snaps 30s NavIC GPS to 1D track chainage KM; RK4 ODE solver computes acceleration & braking.

$$a(t) = [F_{trac}(v) - R_{davis} - F_{grade}] / M_{eff}$$

Machine Learning & Spatio-Temporal Graph (ST-GAT)
Multi-head graph attention models preceding train delays & downstream ripple across 150 km corridor.

$$\alpha_{ij} = \text{Softmax}(\text{LeakyReLU}(a^T [W_h_i \parallel W_h_j]))$$

Statistical Forecasting & Conformal Uncertainty
Certified 90% non-parametric confidence window [P10–P90] adapting to fog halts & TSR speed limits.

Control Room, CIDS & Downstream Logistics APIs
Rule 401 Siding loops freight (+22m recovery); syncs dynamic platform allocation, crew & feeder transport.

MASTER IMPLEMENTATION ARCHITECTURE (INPUT STAGE • PROCESSING STAGE • OUTPUT STAGE)

End-to-End Operational Lifecycle

